

Project Design Phase Proposed Solution Template

Date	1 July 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement	Farmers often struggle to select the most suitable crop because they cannot accurately analyze soil nutrients and environmental conditions, leading to lower productivity and inefficient resource utilization.
2	Idea / Solution Description	Develop a Flask-based web application that uses a trained machine learning model to analyze soil nutrients and environmental parameters and recommend the most suitable crop for farmers.
3	Novelty / Uniqueness	Combines soil nutrient analysis and environmental parameter evaluation using a trained machine learning model deployed through a Flask-based web application to provide accurate and data-driven crop recommendations
4	Social Impact / Customer Satisfaction	Improves agricultural productivity, supports sustainable farming, reduces incorrect crop selection, increases farmer confidence and income.
5	Business Model	Freemium web application for farmers, subscription services for agricultural organizations, licensing for research institutions and government agencies.
6	Scalability	Can be expanded to support additional crops, larger datasets, multiple languages, mobile applications, and integration with future agricultural services.